

CR-46 Internal Measurement Rules and Tolerances

Use these rules only for source-table calculation and status classification. This document is not an approval, acceptance, compliance certification, or redesign instruction.

Rule ID	Quantity / Check	Required Inputs	Formula or Rule
MR-001	Concrete volume	length_m, width_m, thickness_m	length_m x width_m x thickness_m
MR-002	Sidewalk panel area	length_m, width_m	length_m x width_m
MR-003	Driveway apron area	length_m, width_m	length_m x width_m
MR-004	Subbase volume	length_m, width_m, compacted_depth_m	length_m x width_m x compacted_depth_m
MR-005	Detectable warning area	tile_length_m, tile_width_m	tile_length_m x tile_width_m
MR-006	Running slope percent	rise_m, run_m	rise_m / run_m x 100
MR-007	Cross slope percent	crossfall_m, width_m	crossfall_m / width_m x 100
MR-008	Elevation variance	design_elevation_m, asbuilt_elevation_m	asbuilt_elevation_m - design_elevation_m

Tolerance Limits

Tolerance ID	Item	Limit / Band	Status Use
TL-001	Elevation variance	absolute variance <= 0.010 m	Values outside the limit are CONFLICT.
TL-002	Running slope	running slope <= 8.33 percent	Values above the limit are CONFLICT.
TL-003	Cross slope	cross slope <= 2.00 percent	Values above the limit are CONFLICT.
TL-004	Concrete ticket variance	absolute ticket minus calculated volume <= 0.050 m3	Values outside the limit are CONFLICT.
TL-005	Subbase compaction	result >= 95.0 percent	Values below the minimum are CONFLICT.
TL-006	Concrete compressive strength	result >= 30.0 MPa	Values below the minimum are CONFLICT.
TL-007	Concrete slump	75 mm to 125 mm inclusive	Values outside the band are CONFLICT.
TL-008	Detectable warning area	absolute difference <= 0.020 m2 against same tile	Values outside the limit are CONFLICT.

Status Rules

Status	Definition
SOURCE_OK	The source value is directly supported and no conflicting source record is found.
CONFLICT	Another supplied file gives a different value for the same ramp ID, sidewalk panel, driveway apron, station range, pay item, measurement type, and unit.
MISSING	A required source field is blank or absent.
DUPLICATE	The same source repeats the same ramp ID, panel ID, station range, pay item, measurement type, unit, and value.
TEXT_ONLY	The record is descriptive and has no numeric quantity.

- No missing value may be inferred from another file.
- A blank recorded quantity remains MISSING even when a calculated value can be shown from explicit inputs.
- A single source row may produce separate audit lines if it contains separate missing and conflicting facts.
- Do not include payment approval, change-order advice, ADA certification, construction acceptance, completion wording, or sign-off requests.